ADDITIONAL PROBLEMS

1. The number of integers in the range of $f(x)=\frac{1}{\pi}\left(\sin^{-1}(x)+\tan^{-1}(x)\right)+\frac{x+1}{x^2+2x+5}$ **is**

a) 0 b) 3 c) 2 d) 1

Key. c

Sol. $f(x)=\frac{1}{\pi}\left(\sin^{-1}(x)+\tan^{-1}(x)\right)+\frac{1}{(x+1)+\frac{4}{(x+1)}}$

$=g(x)+h(x)$

Here Domain of $g(x)$ is [-1, 1]

$\Rightarrow$ range of $g(x)$ is $\left[\frac{-3}{4},\frac{3}{4}\right]$

Also, max value of $h(x)$ occurs when $(x+1)+\frac{4}{(x+1)}$ is min. at $x=1$

$\Rightarrow$ range of $f(x)$ is $\left[\frac{-3}{4},1\right]$

$\Rightarrow$ Number of integers $=2$

2. If $f(x)$ is continuous function and attains only non-zero rational values, then the number of distinct roots of the eqation $f(0).x^2-6f(3)x+9f(-5)=0$ is

a) 1 b) 2 c) 0 d) ∞

Key. a

Sol. As function attains only rational values, so $f(x)$ is constant function

$\Rightarrow f(0)=f(3)=f(-5)\neq 0$

$\Rightarrow x^2-6x+9=0 \Rightarrow (x-3)^2=0$

$\Rightarrow x=3$

3. the number of functions defined from $f:\{1,2,3,4,5,6\}\to\{7,8,9,10\}$ such that $\sum_{r=1}^{6} f(r)$ is odd, is

a) 2^{10} b) 2^{11} c) 2^{12} d) $2^{12}-1$

Key. b

Sol. Sum can be odd if one (or) three (or) five elements link with odd number

$\therefore\ {}^6C_1.2.2^5+{}^6C_3.2^3.2^3+{}^6C_5.2^5.2$

$=2^6\left({}^6C_1+{}^6C_3+{}^6C_5\right)=2^6.2^5=2^{11}$

4. If $f(x)=\sin^{-1}(x)$ and $g(x)=[\sin(\cos x)]+[\cos(\sin x)]$ then the rage of $f(g(x))$ is, when [.] is G.I.F

a) $\left\{-\frac{\pi}{2},\frac{\pi}{2}\right\}$ b) $\left\{-\frac{\pi}{2},0\right\}$

c) $\left\{0,\frac{\pi}{2}\right\}$ d) $\left\{-\frac{\pi}{2},0,\frac{\pi}{2}\right\}$

Key. d

Sol. $\cos x\in[-1,1]\Rightarrow \sin(\cos x)\in[-\sin 1,\sin 1]$

$\Rightarrow [\sin(\cos x)]=0,-1$

Similarly $[\cos(\sin x)]=0,1$

$g(x)=-1,0,1$

so, range of $f(g(x))$ is $\sin^{-1}(-1)$, $\sin^{-1}(0)$, $\sin^{-1}(1)$

$=\left\{-\frac{\pi}{2},0,\frac{\pi}{2}\right\}$

5. Let $f:R-\{0\}\to[-1,1]$ defined by $f(x)=\frac{e^{x^3}+e^{-x^3}}{e^{x^3}-e^{-x^3}}$, then '$f$' is

a) Many-one function b) one-one function

c) decreasing in $(-\infty,0)$

d) decreasing in $(0,\infty)$

Key. b, c, d

Sol. $f(x)$ is not defined at $x=0$

$$f'(x)=\frac{-12x^2}{\left(e^{x^3}-e^{-x^3}\right)^2}<0,\ x\neq 0$$

6. If $f:R\to\left[\frac{\pi}{6},\frac{\pi}{2}\right)$ defined by $f(x)=\sin^{-1}\left(\frac{x^2-a}{x^2+1}\right)$ is an onto function, then the set of values of 'a' is

a) $(-\infty,-1)$ b) $(-1,\infty)$

c) $(-\infty,0)$ d) $(-\infty,\infty)$

Key. b

Sol. Since $f(x)$ is onto $\frac{\pi}{6}\leq\sin^{-1}\left(\frac{x^2-a}{x^2+1}\right)<\frac{\pi}{2}$

$$\Rightarrow\frac{1}{2}\leq\frac{x^2-a}{x^2+1}<1$$

$$\Rightarrow\frac{1}{2}\leq 1-\frac{(a+1)}{x^2+1}<1,\ \forall x\in R$$

$$\Rightarrow a+1>0\Rightarrow a\in(-1,\infty)$$

Passage: Let $f(x)=x^2-2x-1,\forall x\in R$. Let $f:(-\infty,a]\to[b,\infty)$, where '$a$' is the largest real number for which $f(x)$ is bijective.

Now answer the following.

7. **Let $f:R\to R; g(x)=f(x)+3x-1$ then the least value of the function $y=g(|x|)$ is**

a) $-\frac{9}{4}$ b) $-\frac{5}{4}$ c) -2 d) -1

Key. c

8. **Let $f:[a,\infty)\to[b,\infty)$ then $f^{-1}(x)$ is given by**

a) $1+\sqrt{x+2}$ b) $1-\sqrt{x+3}$

c) $1-\sqrt{x+2}$ d) $1+\sqrt{x+3}$

Key. a

9. **Let $f:R\to R,$ then the range of values of 'K' for which the equation $f(|x|)=K$ has four distinct real roots, is**

a) $(-2,-1)$ b) $(-2,0)$

c) $(-1,0)$ d) $(0,1)$

Key. a

Sol. $f(x)=(x-1)^2-2\Rightarrow a=1;\ b=-2$

$$g(x)=x^2+x-2=\left(x+\frac{1}{2}\right)^2-\frac{9}{4}$$

$$\Rightarrow g(|x|)=\left(|x|+\frac{1}{2}\right)^2-\frac{9}{4}$$

$$g_{\min}=g(0)=-2$$

$$f:[1,\infty)\to[-2,\infty)$$

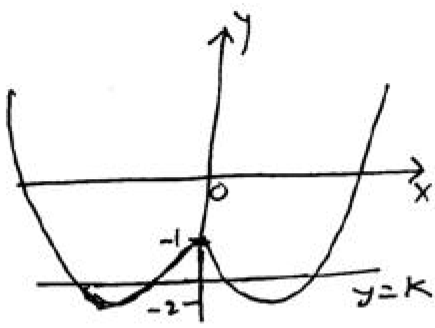
$$f^{-1}:[-2,\infty)\to[1,\infty)$$

$$f(x)=y\Rightarrow x^2-2x-(1+y)=0$$

$$\Rightarrow x=1\pm\sqrt{2+y}\Rightarrow f^{-1}(y)=1+\sqrt{2+y}$$

$$f^{-1}(x)=1+\sqrt{2+x}$$

graph of $y=f(|x|)$



from the graph,

$f(|x|)=K$ has four roots if $K\in(-2,-1)$

10. If a function $f:R\to R$ be such that $f(x-f(y))=f(f(y))+x.f(y)+f(x)-1,$ $\forall x,y\in R$ then $f(2)=$

a) 1 b) 3 c) -1 d) 4

Key. c

Sol. Put $x=f(y)=0$

then $f(0)=f(0)+0+f(0)-1$

$\Rightarrow f(\lambda)=1-\frac{\lambda^2}{2}\Rightarrow f(x)=1-\frac{x^2}{2}$

11. If fractional part of $\frac{1}{x}$ and x^2 for some $x\in(\sqrt{2},\sqrt{3})$ are equal, then the value of $x^4-\frac{3}{x}$ is

a) 3 b) 5 c) 7 d) 0

Key. b

Sol. $\left\{\frac{1}{x}\right\}=\left\{x^2\right\}\Rightarrow\frac{1}{x}=x^2-2$

$\Rightarrow x^3-2x-1=0\Rightarrow(x+1)(x^2-x-1)=0$

where $x=-1\notin(\sqrt{2},\sqrt{3})$

$\Rightarrow x^2-x-1=0\Rightarrow x^4=3x+2$

$\Rightarrow x^4-\frac{3}{x}=(3x+2)-(x^2-2)$

12. If the domain of $f(x)$ is [1, 3] then the domain of $f\left(\log_2\left(x^2+3x-2\right)\right)$

a) $[-5,-4]\cup[1,2]$ b) $[-13,-2]\cup\left[\frac{3}{5},5\right]$

c) $[4,10]\cup[2,7]$ d) $[-3,2]$

Key. a

Sol. $1\le\log_2\left(x^2+3x-2\right)\le3$

$2\le x^2+3x-2\le8$

$-5\le x\le-4$ and $1\le x\le2$

13. Let 'g' be a function defined by $g(x)=\sqrt{\{x\}}+\sqrt{1-\{x\}}$ where {.} denotes the fraction part functio, then g is

a) odd function b) even function

c) one-one function d) many-one function

Key. b, d

Sol. If $x\in Z$ then $\{x\}=0$

$\Rightarrow g(x)=1$

$\Rightarrow g(-x)=g(x)=1$

If $x\notin Z$ jthen $\{-x\}=1-\{x\}$

$\therefore g(-x)=\sqrt{\{-x\}}+\sqrt{1-\{x\}}$

$=\sqrt{1-\{-x\}}+\sqrt{1-(1-\{x\})}$

$=\sqrt{1-\{x\}}+\sqrt{\{x\}}=g(x)$

14. Consider $f:R^+\to R$ such that $f(3)=1$ for $a\in R^+$ and

$f(x).f(y)+f\left(\frac{3}{x}\right).f\left(\frac{3}{y}\right)=2f(xy),$

$\forall x,y\in R^+$, **then** $f(97)=$

a) 1 b) -1 c) 2 d) 97

Key. a

Sol. Put $x=y=1\Rightarrow f(1)=1$

put $y=1\Rightarrow f(x)=f\left(\frac{3}{x}\right),\forall x>0$

$f(x).f\left(\frac{3}{x}\right)+f\left(\frac{3}{x}\right).f(x)=2f(3)$

$\Rightarrow f(x).f\left(\frac{3}{x}\right)=1\Rightarrow f(x)=1\quad(\because x>0)$

15. If the solution set for $f(x)<3$ is $(0,\infty)$ and the solution set for $f(x)>-2$ is $(-\infty,5)$ then the interval in which 'x' lies for the equation $(f(x))^2 \geq f(x)+6$, is

a) $(-\infty,\infty)$ b) $(-\infty,-5]\cup[5,\infty)$

c) $(-\infty,0]\cup[5,\infty)$ d) $[-2,5]$

Key. c

Sol. $f^2(x)-f(x)-6\geq 0$

$\Rightarrow (f(x)-3)(f(x)+2)\geq 0$

$\Rightarrow f(x)\in(-\infty,-2]\cup[3,\infty)$

$x\in(-\infty,0]\cup[5,\infty)$

16. Number of integers in the domain of $f(x)=\log_{[x^2]}(4-|x|)+\log_2\{\sqrt{x}\}$ is (where [.] and {.} have G.I.F, fraction part)

a) 0 b) 1 c) 2 d) 3

Key. c

Sol. $4-|x|>0\Rightarrow x\in(-4,4)$

$[x^2]>0;\ [x^2]\neq 1\Rightarrow [x^2]\geq 2$

$x\in(-\infty,-\sqrt{2}]\cup[\sqrt{2},\infty)$

$\{\sqrt{x}\}>0;\ \sqrt{x}\notin I\ \Rightarrow x=2,3$

17. Let $f(x)=ax^7+bx^3+cx-5$, where a,b,c are constants, if $f(-7)=7$ then $f(7)=$

a) -17 b) -7 c) 14 d) 7

Key. a

Sol. $f(-x)+f(x)=-10$

$\Rightarrow f(-7)+f(7)=-10$

18. If $f(x)$ is a polynomial having integral coefficients such that $f(m)-f(n)=1$, where 'm' and 'n' are integers, then the value of $m-n$, is

a) 2 b) -2 c) 1 d) -1

Key. c, d

Sol. Let $f(x)=a_0+a_1x+a_2x^2+........+a_kx^k$

(where $a_0,a_1,a_2,.....$ are integers)

$f(m)-f(n)=1$

$\Rightarrow a_1(m-n)+a_2(m^2-n^2)+.......$

$a_k(m^k-n^k)=1$

$\Rightarrow (m-n)$ (integers) $=1$

$\Rightarrow (m-n)$ can be 1 (or) -1

acceding as Interger = 1, -1

19. If $f(x)=kx^2-m$ such that $f(1)\in[-4,-1]$ and $f(2)\in[-1,5]$ then the max. value of $f(3)$ is

a) 10 b) 15 c) 17 d) 20

Key. d

Sol. $f(1)=k-m$

$f(2)=4k-m$

$\Rightarrow k=\dfrac{f(2)-f(1)}{3};\ m=\dfrac{f(2)-4f(1)}{3}$

$\Rightarrow f(3)=9k-m=\frac{1}{3}(8f(2)-5f(1))$

$\Rightarrow f(3)\in[-1,20]$

20. The number of paris of integers (x,y) that satisfy $\cos(xy)=x;\ \tan(xy)=y$ is

a) 1 b) 2 c) 4 d) 6

Key. a

Sol. $\dfrac{\sin(xy)}{\cos(xy)}=y\Rightarrow \sin(xy)=xy\Rightarrow xy=0$

$\Rightarrow y=0\ (\because x\neq 0)\Rightarrow x=1$

21. Consider a polynomial $f(x)$ which satisfies

$f(x)=(f'(x))^2$; $\forall x\in R$ then $f(x)$ cab be

a) linear function b) quadratic function
c) a cubic function
d) any polynomial of even degree

Key. b

Sol. $f(x)=a_0+a_1x+a_2x^2+.......+a_nx^n$

on differentiating $f'(x)$ will be of $(n-1)^{th}$ degree

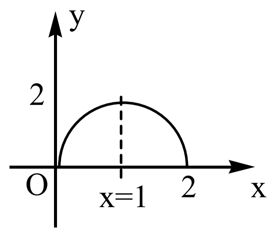
equating degrees $\Rightarrow n=2(n-1)\Rightarrow n=2$

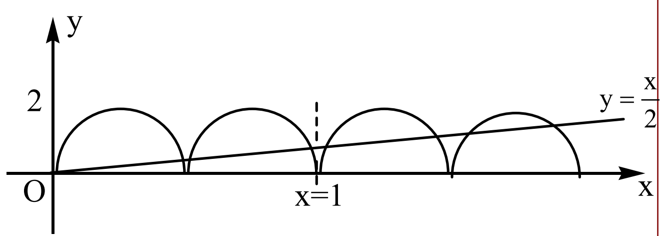
22.. Let $f(x)=2x(2-x)$; $0\le x\le 2$, then the number of solutions of $f(f(f(x)))=\frac{x}{2}$ is

a) 2 b) 4 c) 8 d) 12

Key. c

Sol.





Since the graph $y=f(x)$ is symmetrical about the line $x=1$

Graph of $y=f(f(x))$ and $y=f(f(f(x)))$ are also symmetrical about the line $x=1$

Number of solutions = 8

23. Total number of values of 'x' of the form $\frac{1}{n}, n\in N$ in the interval $x\in\left[\frac{1}{25},\frac{1}{10}\right]$ which satisfy the equation

$\{x\}+\{2x\}+.....+\{12x\}=78x$ (where $\{.\}$ is fractional part function)

a) 12 b) 13 c) 14 d) 15

Key. c

Sol. $(x-[x])+(2x-[2x])+......+(12x-[12x])=78x$

$\Rightarrow [x]+[2x]+.......+[12x]=0$

$\Rightarrow 0\le x<\frac{1}{12}\Rightarrow \frac{1}{25}\le x\le\frac{1}{10}$

$\therefore \frac{1}{25}\le x\le\frac{1}{12}$

24. Number of integers in the range of $f(x)=\frac{1}{\pi}\left(\sin^{-1}x+\tan^{-1}x\right)+\frac{x+1}{x^2+2x+5}$ is

a) 0 b) 3 c) 2 d) 1

Sol. Let $f(x)=g(x)+h(x)$ where domain of $g(x)$ is $[-1,1]$

$\therefore$ Max. of $g(x)=g(1)=\frac{3}{4}$ and min. of $g(x)=g(-1)=-\frac{3}{4}$

Also max. of $h(x)$ occurs when $x+1+\frac{4}{x+1}$ is minimum at $x=1$

$\therefore$ range of $f(x)=\left[-\frac{3}{4},1\right]$

25. Consider $f:R^+\to R$ such nthat $f(3)=1$ for R^+ and $f(x).f(y)+f\left(\frac{3}{x}\right)f\left(\frac{3}{y}\right)$ $=2f(xy)$, $\forall x,y\in R^+$ then $f(97)=$

Sol. Put $x=y=1\Rightarrow f(1)=1$

Now put $y=1$

$$\Rightarrow f(x)=f\left(\frac{3}{x}\right), \forall x>0 \rightarrow (1)$$

$$f(x).f\left(\frac{3}{x}\right)=1 \rightarrow (2)$$

from (1), (2)

$$f(x)=1, \ \forall x>0$$